

Pyrroloquinoline Quinone (PQQ) Potentiates Chemo- and Radiosensitization of Cancer Cells in Synergistic way

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Abstract

Pyrroloquinoline quinone (PQQ), a naturally occurring redox cofactor, has demonstrated a wide range of biological activities, particularly its ability to modulate mitochondrial function and reactive oxygen species (ROS). This study aimed to evaluate the potential of PQQ to enhance the efficacy of chemotherapeutic drugs and radiation in cancer cell lines (A549, MCF-7, and HT-29). Results revealed that PQQ significantly reduced cell viability in a dose-and time-dependent manner, particularly in combination with chemotherapeutic drugs (5-FU, Tamoxifen, Carboplatin, and Doxorubicin) and gamma radiation. Furthermore, PQQ in combination of chemotherapeutic drugs (5-FU) and gamma radiation resulted in increased ROS generation, mitochondrial dysfunction, GSH depletion, and DNA damage, all contributing to enhanced cancer cell death. These findings suggest for first time that PQQ acts as a potent chemo-and radiosensitizer, with the potential to improve cancer treatment outcomes while reducing the required doses of chemotherapy and radiation, thereby possible minimizing side effects.

Introduction

According to the International Agency for Research on Cancer (IARC), the three leading cancers contributing to the global cancer burden are lung, breast, and colorectal cancers. Lung cancer remains the most prevalent, with 2.5 million new cases (12.4%) worldwide, followed by female breast cancer with 2.3 million cases (11.6%), and colorectal cancer in third place with 1.9 million cases (9.6%) during the 2022 [1]. Research in this field is essential to overcoming both economic and psychological burdens of human living with breast cancer, lung and intestine cancer. In this context the cancer cell lines A549 (lung Cancer), MCF-7 (breast cancer) and HT-29 (colon cancer) play a crucial role in this research as they can be used as preclinical models in cancer research.

Pyrroloquinoline quinone (PQQ) is a naturally occurring nutrient that functions as a redox cofactor in bacterial quinoprotein dehydrogenases and kinases, scavenging reactive oxygen species (ROS) [2-7]. Nutritional studies in rodents have shown that PQQ is important for the rodent growth, immune function, blood sugar control and reproductive maturity [8-13]. PQQ has been found in human breast milk [14], but its nutritional benefits for human growth and development remain unclear. Nevertheless, in cultured human and mouse cells, PQQ appears to act as a potential growth factor, promoting cell proliferation when added to culture media [15-17]. Additionally, research shows that PQQ at concentrations below 25-50 μ M protects mitochondria from oxidative stress and exhibits potential in treating or diagnosing neurological and psychological disorders [18-26].

However, studies on mouse cortical neurons indicate that PQQ and its ester derivative (PQQE), at high concentrations, can cause neurotoxicity and apoptosis [27]. Moreover, PQQ has been shown to induce apoptosis in chondrosarcoma cells by increasing intracellular reactive oxygen species (ROS) levels [28], affecting both mitochondrial caspase-dependent and caspase-independent pathways [29]. In U937 cells, PQQ treatment leads to a 2-5-fold increase in apoptosis, primarily due to ROS-induced glutathione depletion, which is a key factor in PQQ's cytotoxic effects [30]. At concentrations above 40 μ M, PQQ can

induce apoptosis in cancer cells, including leukemia and lymphoma. Furthermore, it is suggested that the imbalance in cellular redox homeostasis caused by increased ROS and decreased GSH levels is central to PQQ-induced cytotoxicity [30]. Additionally, this report suggested that cellular redox state plays a critical role in regulating PQQ's effects, with increased GSH levels promoting apoptosis and its depletion potentially leading to necrosis [30].

Given the cytotoxic effects of PQQ, the current study was designed to evaluate its impact in combination with chemotherapeutic drugs (Doxorubicin, 5-FU, Carboplatin, and Tamoxifen) and low-LET gamma radiation on different human cancer cell lines: A549 (lung cancer), MCF-7 (breast cancer), and HT-29 (colon cancer). The results indicated that PQQ acts as a chemo- and radiosensitizer when combined with chemotherapeutic drugs (5-FU, Doxorubicin, Tamoxifen, and Carboplatin) and 2Gy gamma radiation respectively. Furthermore, the combined effects of PQQ with these drugs and gamma radiation were additive. Our first-time report on PQQ's synergistic action with chemotherapeutic drugs and gamma radiation presents a novel approach for delivering effective therapeutic doses at much lower concentrations. This method may potentially minimize the side effects of both chemotherapeutic drugs and radiotherapy, as PQQ allows for reduced effective doses in both treatments.

Results

1. The cell viability of A549 (Lung Cancer), MCF-7 (breast cancer) and HT-29 (colon cancer) suppressed by PQQ

To evaluate the cytotoxic effects of PQQ on A549, MCF-7, and HT-29 cancer cell lines, cells were exposed to varying concentrations of PQQ (0, 10, 20, 40, 80, 160, and 320 μ M) for 24 and 48 hours. Cell viability was measured using the MTT assay, revealing that concentrations above 40 μ M led to a significant increase in cell death compared to untreated controls. This effect became more pronounced as the PQQ concentration increased (Fig. 1). Notably, the cell death was also time-dependent, with greater effects observed at 48 hours compared to 24 hours at concentrations exceeding 40 μ M (compare Fig. 1A vs. Fig. 1B). Furthermore, the data indicated that concentrations of 80 μ M and higher were highly cytotoxic ($p < 0.001^{***}$ or 0.01^{**}) for all tested cell lines. These results suggest that PQQ reduces cell viability in a dose- and time-dependent manner in A549, MCF-7, and HT-29 cancer cells.

2. PQQ cytotoxic effect showed modest synergistic effect with gamma radiation

The cytotoxic effect of an 80 μ M concentration of PQQ was assessed on A549, MCF7, and HT-29 cell lines, both with and without exposure to 2 Gy of gamma radiation, using the MTT assay to measure cell viability. The results showed that PQQ at 80 μ M alone had a cytotoxic effect on A549 and MCF7 cells that was comparable to or greater than the effect of 2 Gy gamma radiation. In contrast, HT-29 cells were more sensitive to gamma radiation than to PQQ alone (Fig. 2A). Furthermore, administering PQQ either 1 hour before (PQQ+2Gy) or after (2Gy+PQQ) gamma radiation led to a similar level of cytotoxicity, with a modest synergistic effect observed when both treatments were combined ($p < 0.01^{**}$) (Fig. 2A). This synergy is noteworthy and suggests that the activation of cellular antioxidant defenses by either PQQ or

gamma radiation may partially mitigate the effect of each treatment. Cell death was further confirmed using propidium iodide (PI) staining, showing that the combination treatment (2Gy+PQQ) induced significant cell death ($p < 0.001^{***}$) in all three cell lines (Fig. 2B). Overall, these findings indicate that while PQQ, particularly in combination with gamma radiation, shows moderate potential as a cytotoxic treatment, its effectiveness may vary across different cancer cell lines.

3. PQQ cytotoxic effect showed strong synergistic effect with chemotherapeutic drugs at their IC₅₀ concentration

Chemotherapeutic drugs, such as Tamoxifen, Carboplatin, 5-FU, and Doxorubicin, are commonly used in clinical cancer treatment. Tamoxifen inhibits the growth of ER+ (estrogen receptor- α -positive) breast cancer cells by blocking estrogen receptors and inducing apoptosis. Carboplatin disrupts DNA replication by forming inter- and intra-strand cross-links, while 5-FU mimics uracil and inhibits thymidylate synthase, thereby blocking DNA synthesis and primarily targeting rapidly dividing cancer cells. Both drugs are widely used to treat a broad range of cancers. Doxorubicin intercalates into DNA, inhibits topoisomerase II, and induces oxidative stress, leading to DNA damage. Despite their varying mechanisms of action, these drugs share a common challenge: they cause toxic effects on normal tissues, including the brain, liver, and kidneys, at therapeutic doses. Thus, achieving the full cytotoxic effect at lower concentration is desirable.

PQQ at a concentration of 80 μ M, has been tested for its synergistic effects with these chemotherapeutic agents. Additionally, cells treated with PQQ and the chemotherapeutic drugs were also subjected to 2Gy gamma radiation to assess the combined effect of PQQ, drugs, and radiation (Fig. 3). The data indicate that all tested drugs, at their IC₅₀ concentrations, exhibited significant cytotoxicity ($P < 0.001^{***}$) in A549, MCF7, and HT-29 cell lines compared to untreated cells. The cytotoxic effect was comparable to that of PQQ (80 μ M) alone ($P < 0.001^{***}$) (Fig. 3). Notably, A549 and HT-29 cells showed a significant reduction in cell proliferation when treated with the combination of PQQ and drugs ($P < 0.001^{###}$) compared to PQQ or the drugs alone ($P < 0.001^{@@@}$) (Fig. 3). In contrast, MCF7 cells showed no significant effect when treated with PQQ and drugs in combination compared to either PQQ / drug treatment alone (Fig. 3). However, both PQQ and the drugs alone caused a significant reduction in cell proliferation compared to untreated cells ($P < 0.001^{***}$). When cells were treated with PQQ, chemotherapeutic drugs, and 2Gy gamma radiation, a significant reduction in cell proliferation was observed in A549, HT-29, and MCF7 cell lines ($P < 0.001^{***}$) compared to untreated cells, PQQ alone ($P < 0.001^{###}$), or drugs alone ($P < 0.001^{@@@}$). Interestingly, all tested cell lines also showed a significant reduction in cell viability when treated with the combination of PQQ, Doxorubicin, and 2Gy radiation compared to PQQ and Doxorubicin alone ($P < 0.01^{\$}$ or $P < 0.001^{$$$}$) (Fig. 3D).

Altogether, the data suggest that chemotherapeutic drugs, which either directly affect DNA metabolism (Carboplatin, 5-FU, and Doxorubicin) or inhibit estrogen signaling (Tamoxifen), significantly reduce cell proliferation when combined with PQQ (80 μ M) at their IC₅₀ concentrations. This effect is further enhanced by gamma radiation (Fig. 3). Additionally, cell viability in response to 5-FU or its combination

with PQQ and gamma radiation is supported by MTT assay results (Fig. S1) and cell death detected by propidium iodide staining (Fig. S2), consistent with the SRB assay for 5-FU (Fig. 3C).

In summary, PQQ (80 μ M) demonstrates potential as a strong synergistic agent in enhancing the efficacy of chemotherapeutic drugs and gamma radiation, significantly reducing cancer cell proliferation. This suggests that PQQ may improve cancer treatment outcomes when used in combination with these therapies.

4. Increased reactive oxygen species (ROS) level coincides with PQQ synergistic effect with 5-FU and gamma radiation treatment

Intracellular ROS generation following treatment with PQQ, or in combination with 5-FU and gamma radiation, was quantitatively assessed using flow cytometry with CM-H2DCFDA dye after 4, 24, and 48 hours of PQQ exposure. As also previously reported, PQQ treatment induces a time-dependent increase in ROS across the tested cell lines, with a significant rise ($P < 0.01^{**}$ for A549 and MCF7, $P < 0.001^{***}$ for HT-29) observed after 48 hours (Fig. 4). However, ROS induction was insignificant in the three cell lines during the first 4 hours of treatment with PQQ or 5-FU alone (Fig. S3A). Notably, ROS levels in cells treated with PQQ alone were similar to those treated with 5-FU ($P < 0.001^{***}$), except in MCF-7 cells, where PQQ induced a lower ROS increase ($P < 0.01^{**}$) compared to 5-FU and gamma radiation (Fig. 4). Additionally, irradiation with 2Gy gamma radiation significantly elevated ROS levels at the 4-hour mark (Fig. S3A), with levels remaining high up to 48 hours (Fig. 4) or staying elevated up to 48 hours in all three cell lines at a concentration of 80 μ M PQQ (Fig. 4 & S3B).

Furthermore, a key observation was that the combination of 5-FU and 2Gy gamma radiation produced ROS levels comparable to the PQQ+5-FU combination, with both treatments leading to a significant increase in ROS ($P < 0.001^{***}$) compared to untreated controls, PQQ alone ($P < 0.001^{###}$), or 5-FU alone ($P < 0.001^{@@@}$) at 4 and 24 hours (Fig. 4 & S3). These levels unchanged by the 48-hour time point (Fig. 4), suggesting that PQQ combined with 5-FU produces a synergistic increase in ROS. Moreover, adding PQQ to the 5-FU + 2Gy treatment resulted in the highest ROS increase across all three cell lines (Figs. 4 & S3). Overall, these results indicate that PQQ enhances intracellular ROS levels and exhibits a synergistic effect when combined with 5-FU and gamma radiation.

5. PQQ showed synergistic effect on reducing the mitochondria membrane potential with 5-FU and gamma radiation treatment

At concentrations of 40-80 μ M, PQQ induces cytotoxicity and exhibits strong anti-tumor activity in A549 and Neuro-2A cells, with minimal impact on normal cells such as HRPTEpiC and HUVEC [31]. The induced apoptosis is concentration-dependent, associated with increased ROS, reduced ATP levels, loss of mitochondrial membrane potential (MMP), downregulation of Bcl-2, activation of caspase-3, and altered MAPK phosphorylation [31]. In this study, MMP loss was assessed with PQQ (80 μ M) alone and in combination with 5-FU or 2Gy gamma radiation for A549, MCF7, and HT-29 cell lines. The data indicated that PQQ alone at 80 μ M moderately reduced MMP in A549 and MCF-7 cells ($P < 0.01^{**}$), while

5-FU alone significantly lowered MMP in all three cell lines at its IC₅₀ concentration ($P < 0.001^{***}$) (Fig. 5). PQQ combined with 5-FU further decreased MMP, nearly matching the effects of 5-FU + 2Gy gamma radiation (Fig. 5). Additionally, the combination of PQQ, 5-FU, and 2Gy gamma radiation resulted in significantly lower MMP compared to untreated cells ($P < 0.001^{***}$), PQQ alone ($P < 0.001^{###}$), or 5-FU alone ($P < 0.001^{@@@}$) (Fig. 5). These findings suggest that PQQ, when combined with 5-FU or gamma radiation, has a strong impact on reducing MMP.

6. PQQ treatment give synergistic effect on reducing cellular GSH with 5-FU

Previous research indicated that PQQ cytotoxic effect is potentiate through the depletion of cellular GSH [30]. In this study, we examined the effect of PQQ alone, as well as in combination with 5-FU, 2Gy gamma radiation, or both, on cellular GSH levels. The results showed that treatment with PQQ, 5-FU, or 2Gy gamma radiation alone did not significantly reduce GSH levels in A549 cells, while modest effects were observed in HT-29 and MCF-7 cells (Fig. 6). However, combining PQQ with 5-FU, or with 5-FU and 2Gy, resulted in a significant reduction in GSH levels ($P < 0.001^{***}$) compared to untreated cells, PQQ alone ($P < 0.001^{###}$), or 5-FU alone ($P < 0.001^{@@@}$) (Fig. 6). Moreover, the combination of PQQ with 5-FU and 2Gy yielded the greatest depletion of GSH levels across all three cell lines (Fig. 6). These findings suggest that PQQ, when combined with 5-FU or gamma radiation, has a potent effect on reducing cellular GSH.

7. PQQ augmented the DNA damage capability of 5-FU and gamma radiation

The extent of DNA strand breaks caused by DNA-damaging agents is directly correlated with their ability to induce cell death. In this study, the degree of DNA strand breaks introduced in A549, HT-29, and MCF-7 cell lines by treatment with PQQ alone or in combination with 5-FU or 2Gy gamma radiation was assessed using the alkaline comet assay. The data indicated that treatment with PQQ (80 μ M) alone did not significantly induce DNA strand breaks, as there was no notable increase in tail length, except in MCF-7 cells, where a moderate increase was observed ($P < 0.01^{**}$) (Fig. 7). Interestingly, treatment with 5-FU or 2Gy gamma radiation alone did not cause substantial DNA strand breaks in any of the cell lines except for MCF-7 (Fig. 7). However, the combination of PQQ with 5-FU significantly increased DNA strand breaks compared to untreated cells ($P < 0.001^{***}$), PQQ alone ($P < 0.001^{###}$), or 5-FU alone ($P < 0.001^{@@@}$), resulting in nearly a three-fold increase in tail length across all tested cell lines (Fig. 7). This effect was further enhanced when PQQ, 5-FU, and 2Gy gamma radiation were combined (Fig. 7). Overall, these findings suggest that PQQ at a concentration of 80 μ M, in combination with 5-FU and 2Gy gamma radiation, is highly effective at inducing DNA strand breaks, ultimately promoting cell death.

Discussion

PQQ, a naturally occurring redox cofactor, has shown promise in a range of biological activities, particularly due to its ability to modulate mitochondrial function and reactive oxygen species (ROS) [8-10, 20, 32-38]. PQQ qualify USA-FDA “generally regarded as safe” (GRAS) requirements and doses up to 4

mg/kg body weight per day or 240 mg/person/day would be safe in adult humans and approved new food ingredient by FDA for the overall health including brain health [39-42]. Further, it is suggested that PQQ effect is concentration dependent and can act as both antioxidant and pro-oxidant [38]. PQQ has a potent antitumor effect and possesses a significant cytotoxic effect on human lung adenocarcinoma, radio- and chemo resistant chondrosarcoma, hepatocarcinoma, promonocytic leukemia, melanoma cells and brain cancer; however, it exhibits little effect on normal cells [28, 30, 31, 43]. The PQQ induces cytotoxic effect on various cancer cell lines by plethora of mechanisms of action, including the induction of reactive oxygen species (ROS) and the depletion of glutathione (GSH), seem to underpin PQQ's potential as an anti-cancer therapeutic and an interesting candidate for cancer therapy as a potential radiation sensitizer [28, 30, 31]. However, practical issue for the uses of PQQ as effective drug for various cancer cell eradication is use of its doses as it is shown that doses between 120-300 μ M showed therapeutic outcome [31, 38]. Thus, it is imperative to achieve the therapeutic outcome of PQQ at lower doses typically in the range of (40-80 μ M). The objective of this study was to evaluate the potential of lower concentration of PQQ (80 μ M) to enhance the efficacy of chemotherapeutic drugs and radiation therapy in cancer cell lines, including A549 (lung cancer), MCF-7 (breast cancer), and HT-29 (colon cancer). The cytotoxic potential of PQQ was assessed in A549, MCF-7, and HT-29 cells, revealing a significant dose- and time-dependent reduction in cell viability. PQQ concentrations above 40 μ M significantly reduced viability, especially at the 48-hour mark (Fig. 1). In combination with 2Gy gamma radiation, PQQ further increased cell death, suggesting a modest synergistic effect (Fig. 2A). This was confirmed through PI staining, showing higher cytotoxicity in the combination treatment compared to either PQQ or radiation alone (Fig. 2B).

When PQQ treatment combined with common chemotherapeutic agents such as 5-FU, Tamoxifen, Carboplatin, and Doxorubicin, PQQ at 80 μ M exhibited a strong synergistic effect on cancer cell proliferation, particularly in A549 and HT-29 cells (Fig. 3). In contrast, MCF-7 cells showed no additional benefit from the combination treatment, though both PQQ and the drugs alone caused significant reductions in cell proliferation. This suggests that the specific cancer cell type plays a role in the sensitivity to PQQ, likely influenced by the differing metabolic and molecular characteristics of each cell line (Fig. 3). The combination of PQQ with chemotherapeutic agents resulted in significant cytotoxicity at their IC_{50} concentrations, suggesting a greater possibility of reducing the required chemotherapeutic drug dosage and has the potential for lowering systemic toxicity. Additionally, when combined with 2Gy of gamma radiation, PQQ and chemotherapeutic agents demonstrated enhanced anti-tumor efficacy in all three cell lines, underscoring PQQ's potential as a radio- and chemo sensitizer.

While the exact mechanism remains to be elucidated, it is likely that PQQ's synergistic action in generating ROS contributes to enhancing the effectiveness of chemotherapeutic drugs and ionizing radiation, which itself induces DNA damage through ROS. The observed synergistic effect of PQQ is likely due to its ability to induce ROS generation and mitochondrial dysfunction. Flow cytometry revealed that PQQ significantly increased ROS levels in all three cell lines in a time-dependent manner (Fig. 4 & S3). When combined with 5-FU or radiation, ROS levels were further elevated, suggesting that the increase in oxidative stress could overwhelm cancer cell defenses, leading to enhanced cell death (Fig. 4

& S3). This was further corroborated by the observed decrease in mitochondrial membrane potential (MMP) following treatment with PQQ, 5-FU, and radiation (Fig. 5). The combination treatments led to significant mitochondrial dysfunction, which is a key indicator of apoptosis. Loss of MMP indicates a failure in the mitochondrial bioenergetic function, often a precursor to cell death via apoptotic pathways [31, 36, 44]. PQQ also demonstrated the ability to deplete cellular glutathione (GSH), a crucial antioxidant in cancer cells [30]. While our results suggested that PQQ alone did not significantly affect GSH levels, the combination of PQQ with 5-FU and gamma radiation led to substantial reductions in GSH (Fig. 6). This depletion of GSH likely contributes to the sensitization of cancer cells to ROS-mediated damage, as GSH plays a critical role in neutralizing ROS and maintaining redox homeostasis. Lower GSH levels reduce the ability of cancer cells to mitigate oxidative stress, enhancing the cytotoxic effects of both PQQ and the combined treatments. Additionally, the combination of PQQ with 5-FU led to a significant increase in DNA strand breaks across all cell lines, suggesting a synergistic effect. This synergism was further amplified when PQQ, 5-FU, and gamma radiation were combined, leading to the most pronounced DNA damage (Fig. 7). These findings indicate that while PQQ alone has limited capacity to induce DNA strand breaks, its combination with chemotherapeutic agents like 5-FU and gamma radiation dramatically enhances DNA damage, which correlates with increased cell death.

Thus, our findings altogether suggest that PQQ, through its ability to generate ROS, reduce mitochondrial membrane potential, and deplete GSH, acts as a potent chemo- and radiosensitizer, enhancing the efficacy of chemotherapeutic drugs and gamma radiation through its synergistic effect. The combination of PQQ with chemotherapeutic agents (Doxorubicin, 5-FU, Carboplatin and Tamoxifen) and ionizing radiation shows promise in reducing cancer cell viability, especially in lung and colon cancer cells. Therefore, current study suggested for the first time that using PQQ as an adjuvant therapy in cancer treatment, potentially allowing for lower doses of chemotherapy and radiation, thereby reducing side effects while maintaining therapeutic efficacy. However, further investigation into the precise molecular mechanisms and in vivo studies are needed to fully understand the therapeutic potential of PQQ in clinical settings.

Abbreviations

3-(4,5-dimethylthiazol-2-yl)-2,5 diphenyltetrazolium bromide (MTT), o-phthalaldehyde (OPT), N-ethylmaleimide (NEM), mitochondrial membrane potential (MMP), reactive oxygen species (ROS), growth inhibition (GI), ionizing radiation (IR), Dulbecco modified Eagle medium (DMEM); propidium iodide (PI), sulforhodamine-B (SRB), Foetal bovine serum (FBS), linear energy transfer (LET).

Materials and Methods

All chemicals used in this study were of analytical grade and were either procured from Indian manufacturers (HiMedia chemicals) or obtained from Sigma-Aldrich and Thermo Scientific. Sulforhodamine B salt (SRB) (HiMedia, cat. no. TC478), MTT [(3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide)] (HiMedia, TC191-1G), CM-H2DCFDA (Thermo Fisher Scientific, C6827),

3,3'-Dipropylthiadicarbocyanine iodide (Sigma, 43608), trypsin, phosphate-buffered saline (PBS) (HiMedia, ML023), agarose, sodium hydroxide (NaOH), EDTA, Tris base, propidium iodide, DMSO, and Dulbecco's Modified Eagle Medium (DMEM) (HiMedia) were used. Fetal bovine serum (HiMedia, RM10432) was also used. Antibodies were procured from Cell Signaling Technology (CST): Phospho-ATR (serine 428) (cat. no. 5428), ub-PCNA lys164 (cat. no. 13439S), and Phospho-Chk1 (S296). Other chemicals used include Tamoxifen (T5648, Sigma), Pyrroloquinoline quinone disodium salt (PQQ) (Methoxatin disodium salt, 80198, Sigma), 5-Fluorouracil (F6627, Sigma), Doxorubicin (DOX) (D1515, Sigma), and Carboplatin (C2538, Sigma).

1. Cell Culture

Human Lung Carcinoma (A549), Human Breast Cancer Cell Line (MCF-7), and Human Colorectal Adenocarcinoma Cell Line (HT-29) were obtained from the National Centre for Cell Sciences, Pune, India. All three cell lines were maintained in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% (v/v) heat-inactivated fetal bovine serum (FBS), penicillin, and streptomycin, with the pH adjusted to 7.4 for optimal growth at 37°C in a humidified atmosphere of 5% CO₂. All experiments were performed using exponentially growing cells, and subculturing was done twice a week. Cells were seeded at 3,000 cells per well in 100 µL of medium in 96-well plates for toxicity experiments, utilizing imaging and fluorescence intensity readout formats, respectively.

2. Irradiation with Cobalt-60 Gamma Radiation

Irradiation of human cancer cells was performed using the Bhabhatron-II Telecobalt unit (Bhabha Atomic Research Centre, Mumbai, India) with ⁶⁰Co (dose rate 1 Gy/min) at ambient temperature. Radiation dosimetry of the unit was carried out by a certified radiation safety officer at the institute, using both the Baldwin Farmer secondary dosimeter and Fricke's chemical dosimeter methodologies. In this experiment, human cancer cells were irradiated with 2Gy of ⁶⁰Co gamma radiation.

3. Cell Treatment

The Human Lung Carcinoma (A549), Human Breast Cancer Cell Line (MCF-7), and Human Colorectal Adenocarcinoma Cell Line (HT-29) were each exposed to a ⁶⁰Co radiation dose of 2Gy. After irradiation, the cells were incubated for 24-48 hours, as required for the subsequent experiments. The cells were treated with five different chemotherapeutic agents: Pyrroloquinoline Quinone (PQQ), 5-Fluorouracil (5-FU), Doxorubicin, Carboplatin, and Tamoxifen, each at their respective IC₅₀ concentrations, for toxicity studies in a sterile laminar flow hood. After determining the IC₅₀ doses, the following concentrations were used in the radiation combination study: PQQ (80 µM), 5-FU (5 µM), Doxorubicin (70 µM), Carboplatin (5 µM), and Tamoxifen (19.4 µg/ml), with a 1-hour pretreatment before radiation exposure.

4. SRB-uptake assay for cellular toxicity

The SRB uptake assay to measure cellular toxicity was performed as described previously [45]. Exponentially growing (3×10^3) cells (A549, MCF-7, and HT-29) were seeded into 96-well plates and incubated to allow attachment. After various treatments and incubations, as outlined in the treatment protocol, cytotoxicity was assessed based on cell proliferation. Briefly, following the treatment and incubation, cells were washed with phosphate-buffered saline (PBS), fixed with 10% trichloroacetic acid (w/v) for 1 hour at 4°C to precipitate proteins, then washed with deionized water and stained with 0.4% SRB (w/v) in 1% acetic acid (100 µl/well) for 1 hour at room temperature. The plates were washed 4-5 times with deionized water to remove unbound dye and air-dried at room temperature. The SRB-bound dye was extracted using 100 µl of 10 mM Tris-base buffer (pH 10.5), and the absorbance was measured at 565 nm with a reference wavelength of 690 nm using a 96-well multimode plate reader (BioTek Synergy H1 Hybrid Multi-Mode Microplate Reader).

5. MTT assay

MTT assay was used to measure metabolic viability of cell after various treatments as described elsewhere [46]. (3×10^3) cells were seeded in 96-well plates to assess their sensitivity to inhibitors. 5 mg/mL stock solution of MTT was prepared, followed by incubation for 4 hours at 37°C. DMSO was then added, and the optical density (OD) was recorded at 570 nm (Victor) after 48 hours of incubation. Viable cells produced a dark blue formazan product, while no such staining was observed in dead cells. The percentage of viable cells was calculated using the following formula: (%) = $[100 \times (\text{sample absorbance}) / (\text{control absorbance})]$.

6. Estimation of Glutathione

Glutathione was determined by using the method of Hissin & Hilf (1976) [47]. Briefly, 0.6×10^6 A549, HT-29, and MCF-7 cells were grown in 60 mm cell culture petri dishes (70–80% confluency). After various treatments, the cells were trypsinized and washed with ice-cold 1xPBS. A 0.5 ml cell lysate was suspended in 1 ml of 25% HPO₄ and 3.75 ml of sodium phosphate buffer (0.1 M, pH 8.0, with 5 mM EDTA). The samples were centrifuged at 10,000 rpm for 10 minutes at 4°C, and the supernatant was collected for the measurement of total glutathione (GSX) and oxidized glutathione (GSSG). 100 µl aliquot of the supernatant was incubated with 100 µl of o-phthalaldehyde (0.1% in methanol) and 1.8 ml of 0.1 M phosphate buffer (pH 8.0) for 15 minutes at room temperature in the dark. Fluorescence was measured using a fluorescence spectrophotometer (Varian, USA) at an excitation wavelength of 350 nm and an emission wavelength of 420 nm. For the estimation of GSSG, 40 µl of 0.04 M N-ethylmaleimide was added to the supernatant, and the solution was incubated for 30 minutes at room temperature. The same procedure was followed, except that 0.1 M NaOH was used in place of the phosphate buffer. GSSG concentration was measured using a calibration curve.

7. Comet assay

Alkaline Comet assay was performed to determine double strand DNA breaks as described previously [45]. Standard frosted microscope slides were dipped in 1% normal melting point agarose

(NMPA). The excess was wiped from the back and the slides were air dried overnight, they were then stored in a microscope box. Prior to cell harvesting, we prepared alkaline lysis buffer (2.5 M NaCl, 100mM EDTA (pH 8.0), 10 mM Tris-HCl (pH 7.6), 1% Triton X-100, pH to 10), alkaline electrophoresis buffer (200mM NaOH, 500 mM EDTA, pH 13) and alkaline unwinding buffer (200mM NaOH, 200mM EDTA, pH 13) were prepared and chilled to 4°C. Briefly, following various treatment (1×10^3) cells were harvested, mixed with low melting agarose for layering on agarose pre-coated slides. After gel formation slides were kept in lysis buffer (40min at 4°C). After lysis samples were transferred to unwinding buffer and alkaline electrophoresis buffer. Thereafter, samples were dehydrated (70% ice cold ethanol) and air dried (overnight in dark). Slides were stained with propidium iodide (5µg/ml) in dark and examined by fluorescence microscopy.

8. Measurement of ROS

Measurement of Total ROS levels. The changes in cellular ROS levels were measured using CMH₂DCFDA dye by 96 well plate reader, as described elsewhere [45]. Briefly, 1×10^6 cells were plated in 60mm cell culture dishes for each condition, and after treatments, the cells were trypsinized, harvested and incubated with CM-H₂DCFDA (10µM) at 37°C for 30min in dark. Following incubation, the fluorescence of oxidized dye was acquired at λ_{Ex} 488nm, λ_{Em} max 530 nm from at least 10,000 cells/sample. Measurement was done by fluorescence 96 well (BioTek Synergy H1 Hybrid Multi-Mode Microplate Reader), Results are expressed as mean fluorescence.

9. Measurement of cell death by propidium iodide

Cell death measurement by propidium iodide (PI) dye measured as discussed elsewhere [48]. PI staining was used as an indicator of cell death, binding to the major groove of DNA but unable to passively traverse cells with an intact plasma membrane. 5×10^5 cells/ml were harvested from the 60mm culture dish after various treatments as described in the treatment protocol. The cells were washed with PBS and trypsinized to remove contaminants. A concentration of 0.5 µg/ml PI was added to the cell suspension, which was then incubated at room temperature (37°C) for 10-15 minutes. Fluorescence measurements were conducted using a 96-well plate (BioTek Synergy H1 Hybrid Multi-Mode Microplate Reader) at λ_{Ex} 535nm/ λ_{Em} 615nm. PI staining (red fluorescence) was used to quantify cell death. Cell death was determined by subtracting the background (unstained) cell population from the dead cell population.

10. Changes in Mitochondrial Membrane Potential

The measurement of mitochondrial membrane potential (MMP) was carried out as previously described [48]. A549, HT-29, and MCF-7 cells (1×10^5 cells) were seeded in 60 mm cell culture dishes and treated according to the protocol outlined in the treatment section. After various treatments, the cells were washed with 1xPBS and incubated with DiSC3(5) at a working concentration of 2 µM at 37°C for 10 minutes in the dark. Changes in DiSC3(5) fluorescence were measured using a fluorescence

spectrophotometer. Mitochondrial membrane potential was determined using a black 96-well plate reader (BioTek Synergy H1 Hybrid Multi-Mode Microplate Reader) with 3,3'-Dipropylthiadicarbocyanine iodide (Sigma-43608) (λ_{Ex} 500 nm and λ_{Em} 700 nm).

11. Statistical analyses

Data are expressed as mean $\pm$ SEM. All experiments were independently performed at least three times. Statistical analysis was conducted using two-way ANOVA (analysis of variance), followed by post hoc analysis with Bonferroni post-tests (Prism 5.0, GraphPad), unless otherwise stated. All columns were compared across different groups.

Declarations

Conflict of Interest: Authors have no conflict of interest in the content of the manuscript.

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Author's contributions

YSR-conceived research hypothesis, supervised, data analysis, manuscript-writing editing, figure preparation, and final compilation. ML-designed the experiments, performed the experiments, data analysis, figure preparation, and manuscript editing. All authors read, reviewed and approve the final manuscript.

Data Availability statement

The authors confirm that the data supporting the findings of this study are available within the article or its supplementary materials. However, specific data that support the findings of this study are available from the corresponding author, (Email: ysraj@barc.gov.in), upon request.

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Figures

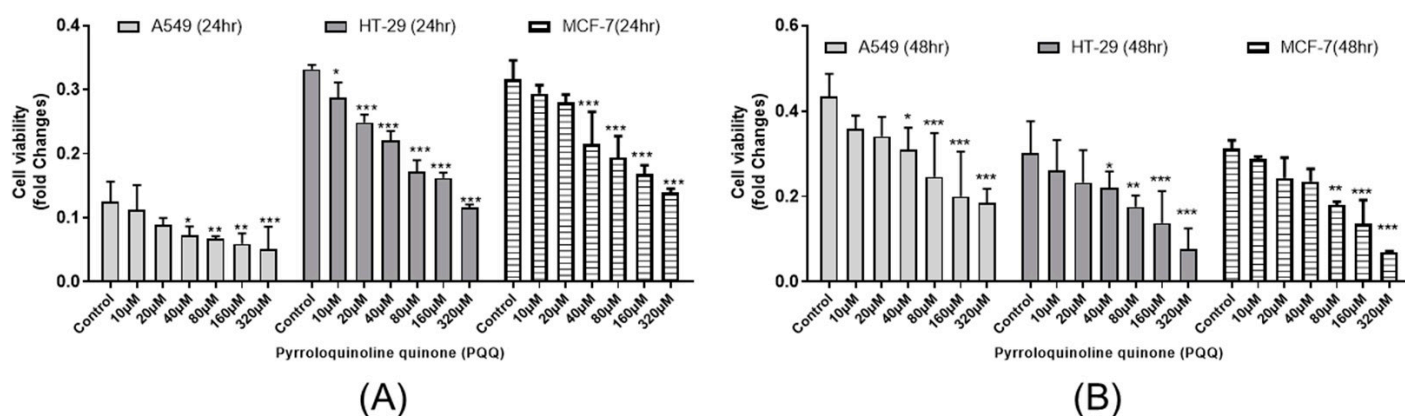


Figure 1

Figure 1

Cell viability assessed by MTT assay of A549, HT-29 and MCF-7 cells treated with PQQ (10-320 μ M); at 24 and 48hrs. (A) Dose-dependent effect on cell viability at 24 hrs. (B) Dose dependent decrease in cell population was also noted at 48hrs. Data shown are representative of three independent experiments and values are given as mean $\pm$ SD. significant difference was observed in cell viability among the different dose groups. Cells treated with 10-320 μ M of PQQ. Statistical analysis was performed by two way analysis of variance (ANOVA) with all pairwise multiple comparison procedures done followed by post hoc analysis using Bonferroni post-tests (Prism 5.0), GraphPad (all columns were compared to different groups) unless otherwise stated. Differences were significant at values ($P < 0.05^*$, $P < 0.01^{**}$, $P < 0.001^{***}$) vs control.

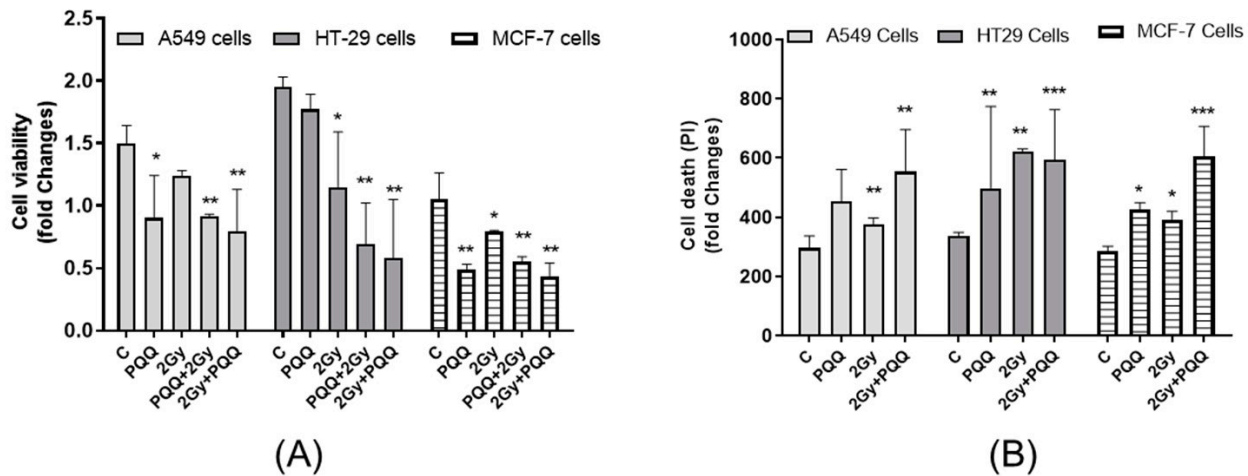


Figure 2

Figure 2

(A) Cell viability assessed by MTT assay of A549, HT-29 and MCF-7 cells treated with PQQ (80 μ M) alone or in combination with 2Gy gamma radiation where PQQ treatment to cells done 1 hour before gamma irradiation (PQQ+2Gy) or after gamma irradiation (2Gy+PQQ). (B) Cell death measure by propidium iodide (PI) staining measured with microplate reader. Data shown are representative of three independent experiments and values are given as mean $\pm$ SD. Statistical analysis was performed by two way analysis of variance (ANOVA) with all pairwise multiple comparison procedures done followed by post hoc analysis using Bonferroni post-tests (Prism 5.0), GraphPad (all columns were compared to different groups) unless otherwise stated. Differences were significant at values ($P < 0.05^*$, $P < 0.01^{**}$, $P < 0.001^{***}$) vs control.

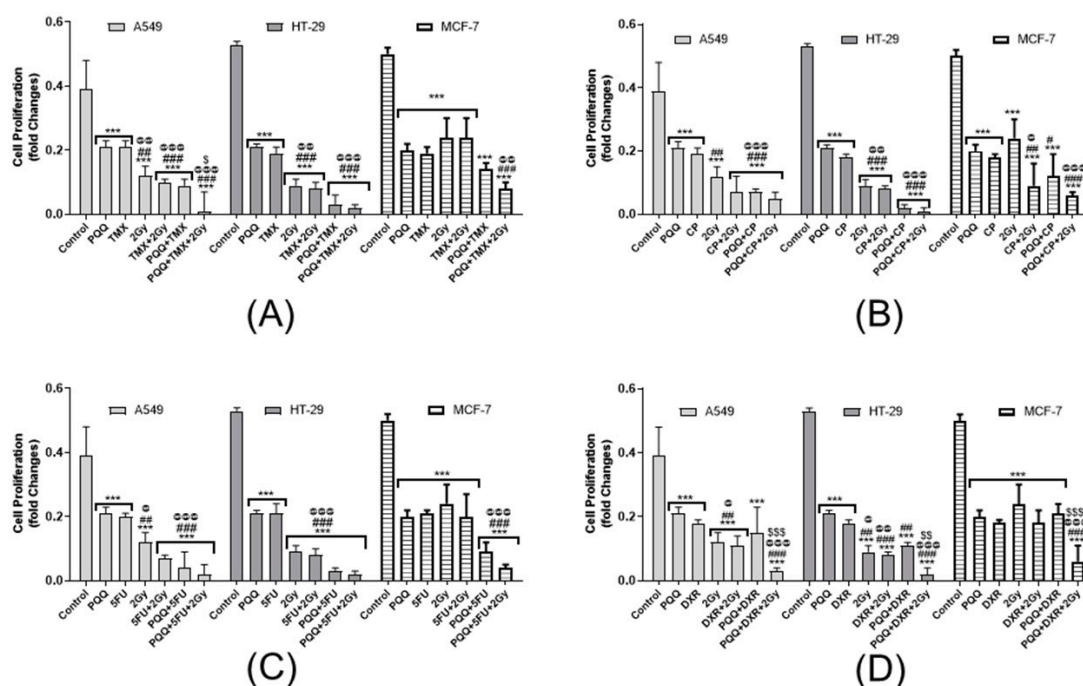


Figure 3

Figure 3

Sulforhodamine B uptake Assay. Effects of cell treatment of A549, HT-29 and MCF-7 cells (48hrs), synergistic action of PQQ (80 μ M) with (A) Tamoxifen (TMX); concentration 19.4 μ g/ml (B) Carboplatin (CP); concentration 5 μ M (C) 5-FU; concentration 5 μ M, and (D) Doxorubicin (DOX); concentration 70 μ M in alone or in combination with gamma radiation (2Gy) were measured in terms of cell proliferation (Sulphorhodamine-B uptake assay) 48 hours. Results are expressed as cell proliferation (fold changes) with respect to (Control vs different treatment*, PQQ alone vs different treatment #, drug alone vs different treatment @ and PQQ+drug vs PQQ +drug+2Gy\$), mean $\pm$ SD of three independent experiments. Differences were designated significant at values $P < 0.05^*$, $P < 0.01^{**}$, $P < 0.001^{***}$ and were labelled with asterisks (*, #, @, \$) and non-significant (ns). Statistical analysis of data was done using two-way (ANOVA) (analysis of variance) followed by post hoc analysis using Bonferroni post-tests (Prism 5.0), GraphPad (all columns were compared to different groups) unless otherwise stated.

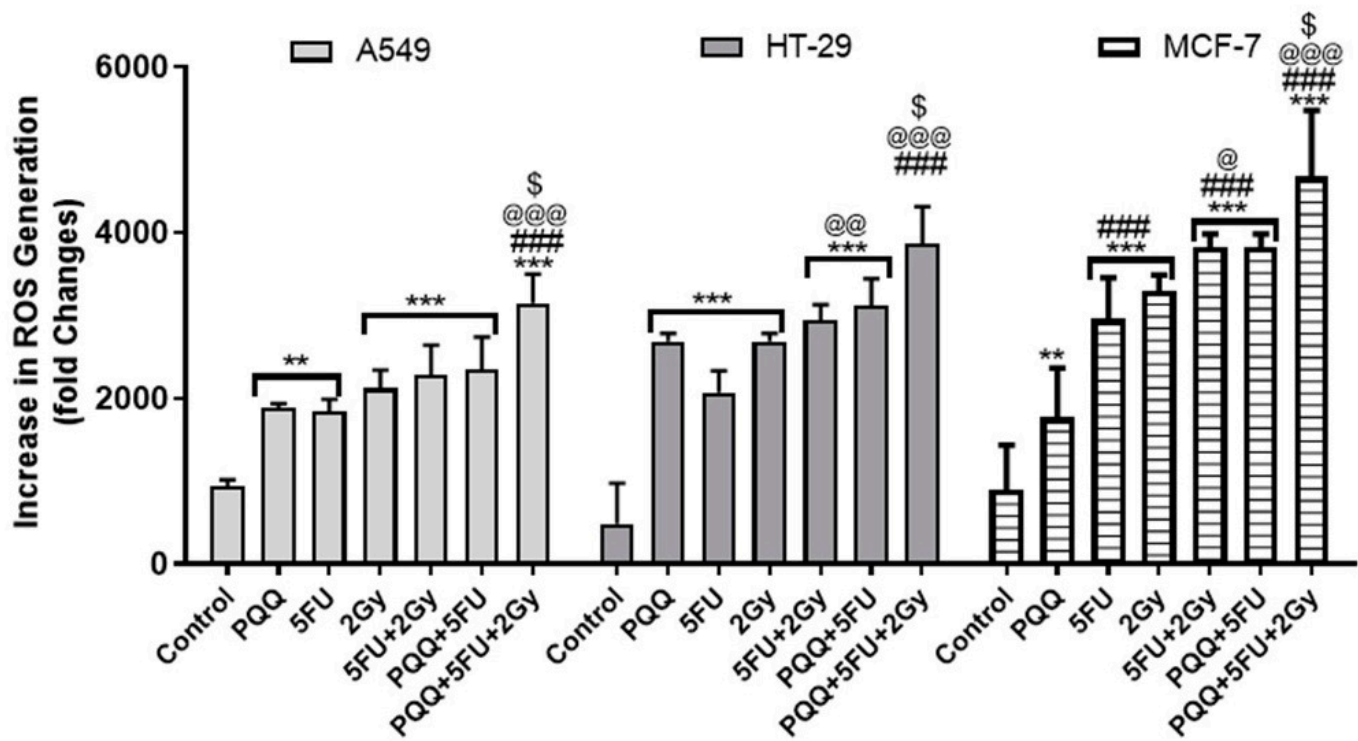


Figure 4

Figure 4

Dose dependent increase of ROS generation in A549, HT-29 and MCF-7 cells after various treatment. ROS mediated oxidation of CM-H₂DCFDA and measurement of the fluorescent CM-DCF with time (48 hours) as an indicator of redox status of cells. Briefly, cells treated with PQQ, 5-FU, and 2Gy gamma radiation alone or PQQ in combination with 5-FU and 2Gy gamma radiation were harvested and thereafter oxidation of dye was measured using λ_{Ex} 488 and λ_{Em} 530 as described under materials and methods. Obtained results presented are representative of three independent experiments. Comparison was done among (Control vs different treatment*, PQQ alone vs different treatment#, drug alone vs different treatment@ and PQQ+drug vs PQQ+drug+2Gy\$). Statistical analysis was performed by two way analysis of variance (ANOVA) with all pairwise multiple comparison procedures done by post hoc analysis using Bonferroni post-tests (Prism 5.0), GraphPad (all columns were compared to different groups) unless otherwise stated. Differences were significant at values $P < 0.05^*$, $P < 0.01^{**}$, $P < 0.001^{***}$, and were labelled with asterisks (*, #, @, \$) and non-significant (ns). Results are representative of mean fluorescence and expressed as fold changes.

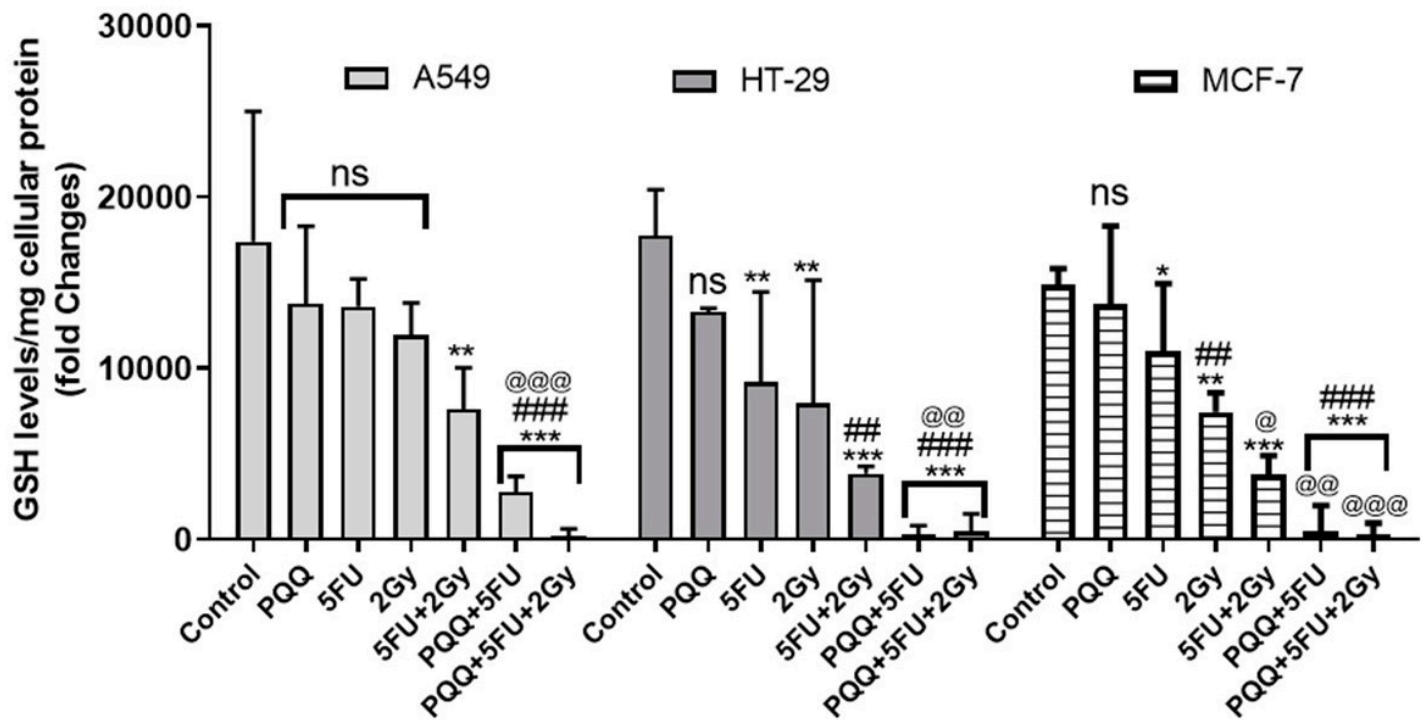


Figure 6

Figure 6

Effect of PQQ and 5-FU treatment of A549, HT-29 and MCF-7 cells on radiation-induced changes in levels of glutathione at 48 hours. The results have been expressed as levels of glutathione/mg of cellular protein. The significance between different groups was analysed using 2-way ANOVA, (fold changes) with respect to (Control vs different treatment*, PQQ alone vs different treatment#, drug alone vs different treatment@ and PQQ+drug vs PQQ+drug+2Gy\$), mean $\pm$ SD of three independent experiments. Statistical analysis of data was done using two-way (ANOVA) (analysis of variance) followed by post hoc analysis using Bonferroni post-tests (Prism 5.0), GraphPad (all columns were compared to different groups) unless otherwise stated. Differences were significant at values $P < 0.05^*$, $P < 0.01^{**}$, $P < 0.001^{***}$ and were labelled with asterisks (*, #, @, \$) and non-significant (ns).

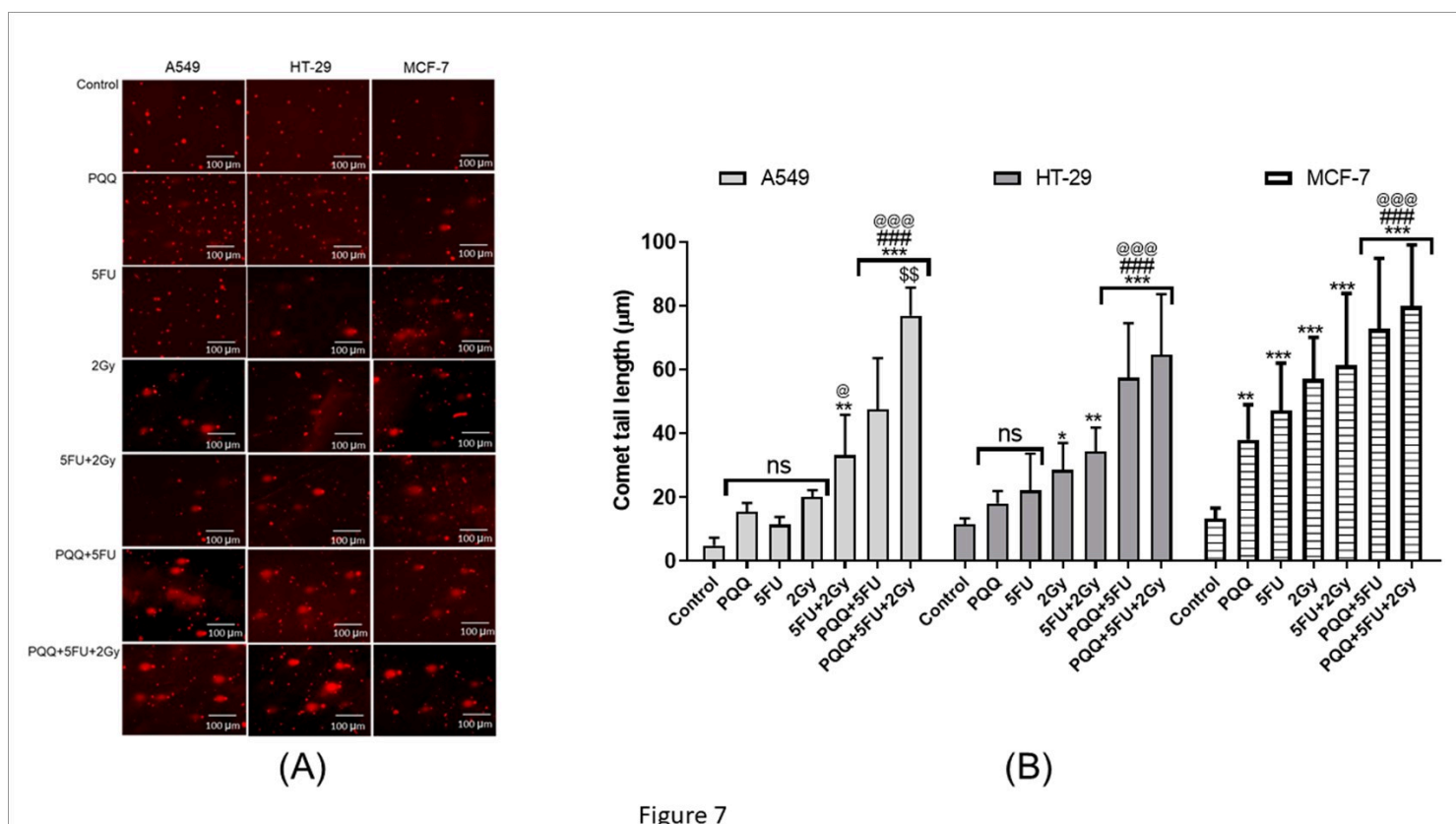


Figure 7

Figure 7

Detection of DNA damage by alkaline comet assay / single cell electrophoresis (SCGE). Comet assays for A549 cells, HT-29 cells, and MCF-7 cells were performed under alkaline conditions to determine the amount of double-strand DNA breaks (48 hours). Slides were stained with PI (1 μg/ml) and examined by fluorescence microscopy, as described in the methods section. Each experiment performed three times after various treatment condition. (A) Representative image of three independent experiment for the various treatment condition. (B) Results are expressed as comet tail length mean ± SD of three independent experiments (TL) with respect to various treatment (48 hours). Comparison was done among (Control vs different treatment*, PQQ alone vs different treatment[#], drug alone vs different treatment[@] and PQQ+drug vs PQQ +drug +2Gy^{\$}), Differences were significant at values P< 0.05*, P<0.01**, P<0.001***, and were labelled with asterisks (*, #, @, \$) and non-significant (ns). Statistical analysis of data was done using two-way (ANOVA) (analysis of variance) followed by post hoc analysis using Bonferroni post-tests (Prism 5.0), GraphPad (all columns were compared to different groups) unless otherwise stated.

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